

# Standard Operating Procedure (SOP)

Preparation, Review, Administration, and Publication of PMC Examination Papers

Sundar STEM School

## 1 Purpose & Principles

This SOP defines the quality assurance process for designing, reviewing, approving, and grading Pakistan Math Contest (PMC) examination papers. Because the PMC is a national-level selection mechanism for Sundar STEM School, all examinations must be academically rigorous, fair, and entirely free of ambiguities or typographical errors. Additionally, to ensure transparency and provide authentic preparation resources, the school mandates the publication of the examination paper, answer key, and detailed solutions after every testing cycle.

## 2 Examination Design & Structure

### 2.1 Assessment Philosophy

The PMC is designed to measure genuine mathematical aptitude. Problems must evaluate a student's capacity to execute mathematical procedures, make sound logical extensions of ideas, perform multi-level analysis, and identify patterns.

The contest is not designed to trick students into selecting the wrong answer. Problems must avoid artificial difficulty: there should be no hidden meanings, semantic traps, or crucial information buried in fine print. The wording of every problem must be as clear and transparent as possible, ensuring that the challenge lies entirely in the mathematical resolution, not in deciphering the question's intent.

To ensure the PMC measures reasoning ability rather than language proficiency, all problems must maintain a low language load. Sentences should be short, and figural or numeric items should be preferred over verbally heavy ones.

**Options Construction** For multiple-choice items, wrong options must correspond to predictable errors rather than random numbers.

### 2.2 Test Blueprint

Before drafting an examination, Subject Matter Experts (SMEs) must create a test blueprint that ensures major curriculum topics, logic applications, and pattern recognition are proportionally represented according to the assessment philosophy.

### 2.3 Difficulty Tiers

- **Medium:** Challenging problems derived from the Single National Curriculum.
- **Hard:** 1-Star or 2-Star level problems (AoPS/Beast Academy standard).
- **Very Hard:** Problems requiring highly creative, outside-the-box thinking.

*(Note: SMEs may recalibrate these definitions based on historical PMC performance data).*

## 2.4 Difficulty Distribution

- **Level 1 Exams:** Distributed evenly (approximately 33% Medium, 33% Hard, 33% Very Hard) to assess a broad spectrum of abilities.
- **Level 2 Exams:** Designed for screening-stage qualifiers, consisting exclusively of 50% Hard and 50% Very Hard questions.

## 2.5 Organization

To minimize cognitive switching, questions must be grouped logically by type (e.g., Mathematical vs. Logic/Pattern-recognition) and style. Within each section, questions should sequence progressively from easier to more difficult.

## 3 Preparation & Independent Review

The Examination Author is responsible for drafting the examination paper alongside an answer key and a solution manual.

Before final approval, the complete examination package must undergo an independent review. The reviewer, uninvolved in the drafting process, must attempt the paper under examination conditions to verify accuracy, appropriate difficulty, and adherence to the test blueprint. The reviewer should also flag any questions which take considerably longer than the intended time or have an alternative defensible answer.

## 4 Grading, Publication & Continuous Improvement

During the evaluation phase, graders must record individual question scores alongside the student's total score.

Following the declaration of results, Sundar STEM School will publish the examination paper, the answer key, and the solution manual. Finally, the academic team will conduct a post-cycle review of the examination statistics (e.g., question success rates) to refine and improve the quality of subsequent PMC exams.